Causal Alignment Prompting (CAP) for Synthetic Survey Augmentation in Policy-Framing RCTs

Pre-Analysis Plan (PAP)

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Abstract

We test whether large language models (LLMs) can act as validated synthetic survey respondents to augment small randomized experiments *without* introducing systematic bias. We propose *Causal Alignment Prompting (CAP)*, a principled augmentation protocol aligning (i) marginal persona priors (demographic strata), (ii) conditional response distributions $p(Y \mid X, T)$ via temperature tuning and rejection sampling, and (iii) effect-direction consistency gleaned from a small human pilot; this soft constraint never hard-codes effect magnitudes. We implement CAP in a policy-framing RCT (gain vs. loss) with $N = 300$ human participants (stratified randomization on education $\times$ gender $\times$ ideology), and generate CAP-aligned synthetic respondents per persona $\times$ arm. Synthetic respondents enter analysis down-weighted by $\omega \in \{0, 0.2, 0.35, 0.5, 0.7\}$, with $\omega = 0$ the confirmatory estimand and $\omega > 0$ precision-oriented under stringent validation checks. Primary outcome is 7-point policy support; secondary outcomes are donation willingness (binary) and a 0–5 knowledge score. We preregister missing-data handling (MICE), non-compliance (ITT primary; CACE via IV), and attrition (IPW), and we fully version prompts, seeds, and CAP settings for transparency and reproducibility. This PAP incorporates detailed validation, multi-model bias checks, weight-interpretation rules, expanded stratification for HTE, and an explicit small-pilot design.

Registry (planned): AEA RCT Registry (ID/link to be added upon filing)
Materials repository (planned): OSF/GitHub with versioned prompts, seeds, CAP configs
IRB/Ethics: To be filed prior to data collection; consent language includes AI-assisted analyses.

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1. Introduction

Research Question. Can large language models (LLMs) serve as validated *synthetic survey respondents* that augment small randomized experiments *without* introducing systematic bias, while *increasing* statistical precision? We study this in a policy-framing RCT (gain vs. loss wording), a setting with tight samples, meaningful heterogeneity, and a premium on transparent, reproducible protocols.

Why This Matters. Surveys are costly and underpowered for moderate effects or HTE. LLMs can generate attribute-conditioned responses quickly and cheaply. If aligned without distorting causal estimates, they enable faster prototyping, stronger design diagnostics, and richer robustness/HTE. If alignment fails, documenting failure modes clarifies responsible use and boundary conditions.

Conceptualization. Synthetic responses are draws from a generative process (model, version, parameters). They are used only after transparent *alignment and validation* with human data. The confirmatory estimand remains the human-population ATE; augmented estimates are interpreted as mixture-population estimands that must be equivalent to human-only within a preregistered margin.

Methodological Contribution. Causal Alignment Prompting (CAP) aligns (i) persona priors, (ii) conditional response distributions $p(Y \mid X, T)$ via temperature tuning and rejection sampling, and (iii) direction-of-effect coherence using a small human pilot (no magnitude hard-coding). Prompts, seeds, versions, and parameters are frozen and versioned. We also propose clear validation checks (pattern replication, effect-direction coherence, external benchmarking) and multi-model sensitivity.

Empirical Fit and Overview. We run a two-arm textual RCT with standard survey outcomes (support, donation, knowledge) and stratified randomization. Human-only analysis is confirmatory; validated synthetic respondents may be included with down-weights $\omega \in \{0.2, 0.35, 0.5, 0.7\}$ chosen by a preregistered rule trading off bias and precision.

2. Hypotheses

Let $T \in \{\text{gain}, \text{loss}\}$ and outcomes Y_1 (primary), Y_2, Y_3 (secondary).

H1. Average Treatment Effect (ATE): Gain vs. loss framing changes mean support $E[Y_1]$. Direction is a priori ambiguous; two-sided.

H2. Heterogeneous Effects (HTE): ATE varies by ideology, education, and gender (primary strata); by age, income, and region (secondary).

H3. Augmentation Validity: With $\omega \in \{0.2, 0.35, 0.5, 0.7\}$, augmented ATEs are equivalent to human-only $\omega = 0$ within ± 0.15 SD and maintain nominal size in placebos.

Exploratory Hypotheses (non-confirmatory).

- H4: Knowledge (Y_3) mediates part of the framing effect on support (Y_1); we will report mediation descriptively without causal identification claims.
- H5: Effects are larger among low-knowledge respondents at baseline; tested as an interaction in exploratory models.

3. Population and Sample

Target Population and Eligibility

Target population: U.S. adults (18+), English-proficient. Inclusion criteria: consent, age $\geq$18, completed attention check(s) on first attempt. Exclusion criteria: duplicate submissions (platform ID/device/IP), incomplete key outcomes, manifest low-effort (e.g., straightlining).

Sampling Frame and Provision of Data

For this course project, the instructor will provide pilot and main datasets designed to approximate an online-panel draw consistent with our stratification quotas. Pilot and main are non-overlapping by construction.

Sample Size and Power

Human sample $N = 300$. Detectable standardized effect $\approx 0.22{-}0.25$ SD at $\alpha = .05$, $1{-}\beta = .80$ under balanced stratified design with modest intra-stratum variance share ρ. We report an MDE table by outcome and (if feasible) primary strata.

Strata Feasibility and Merging Rule

Maintain ≥ 15 humans per occupied primary stratum$\times$arm. If sparse: merge to nearest ideology bin; if still sparse, merge by education; all merges documented.

Synthetic Size and Use

For each human in a stratum×arm, generate $k = 3$ synthetic respondents. Synthetic are included only after all validation checks pass (Sections 5 and 8).

4. Treatment and Randomisation

Treatment Materials

Two versions of the policy description (gain vs. loss framing) differing only in wording. Exact text is frozen and versioned (hashes archived).

High-Level Assignment Plan

Stratified randomisation with permuted blocks (sizes 4/6 varied) and a fixed RNG seed to ensure balance within each education×gender×ideology cell. Assignment list and seed are archived prior to analysis.

CAP Prompts and Settings

Persona prompt encodes demographic priors (stratum membership); task prompt presents items under assigned arm T. Alignment uses a temperature grid $\{0.2, 0.7, 1.0\}$ and rejection sampling to meet pilot-derived moment bands. All prompts, seeds, and parameters are versioned for reproducibility.

5. Threats to Validity

Validation of Synthetic Respondents

Pattern Replication (Human vs. Synthetic). Using the pilot hold-out (Section 8):

1) *Means:* standardized differences < 0.10 within strata×arms for $Y_1 - Y_3$.
2) *Demographic gradients:* reproduce signs and rank ordering (e.g., ideology vs. support).
3) *Variances:* within-stratum ratio $s^2_{\text{syn}}/s^2_{\text{hum}} \in [0.75, 1.25]$; if outside, increase temperature or relax rejection band and re-generate.
4) *Distributions:* two-sample KS (ordinal-as-rank) and Mann–Whitney robustness.

Effect-Direction Coherence. Pilot ATE sign is a soft constraint: in 1,000 bootstrap resamples, the synthetic effect direction must not contradict pilot direction in $> 25\%$ draws.

External Benchmarking. Compare pooled human vs. synthetic to external benchmarks on related constructs; require standardized differences < 0.15 on overlaps.

Additional Risks and Mitigations

- *Construct validity.* We freeze wording and use attention checks; knowledge items are pretested for clarity.
- *Measurement variance distortion.* We explicitly validate dispersion and adjust generation temperature as needed.
- *Model-specific bias.* See multi-model checks below; we disclose model/version/date/parameters and treat model-specific discrepancies as sensitivity only.

Model-Specific Bias and Multi-Model Sensitivity

We pre-specify three LLMs (US/EU/non-US variants). Run CAP and validation for each. A *Model Set Pass* requires ≥ 2 models to pass all checks; one pass $\Rightarrow$ augmented estimates labeled *sensitivity-only*; none $\Rightarrow$ human-only.

6. Outcomes, Diagnostics, and Covariates

Outcomes

Primary: Y_1 7-point support. Secondary: Y_2 donation (0/1), Y_3 knowledge (0–5).

Table 1: Outcomes and Measurement

Outcome	**Type**	**Question/Construction**	**Primary/Secondary**
Y_1 Policy Support	Ordinal (1–7)	7-point Likert on policy support (framing varies by arm)	**Primary**
Y_2 Donation	Binary (0/1)	Willingness to donate $10 to an advocacy group (Yes/No)	Secondary
Y_3 Knowledge	Integer (0–5)	Sum of 5 factual items about the policy (each scored correct/incorrect)	Secondary

Notes: Exact wording of stimuli and items is frozen prior to data collection and archived with commit hashes. Attention checks reported but not used as outcomes except in placebo tests; reverse-coding (if any) is documented in the dictionary.

Diagnostics

We report: (i) arm balance within strata (table), (ii) distribution plots for $Y_1 - Y_3$ by arm and strata, (iii) design effects, and (iv) failure counts (if any) for validation thresholds with rationale.

Covariates

Confirmatory models include stratum fixed effects; ancillary specifications may include age (3), income (3), region (4). If baseline/pre-treatment items exist, they appear only in ancillary models alongside preregistered core models.

7. Randomisation

Operational Procedure

Within each primary stratum we use permuted blocks (sizes 4/6 varied) with a fixed RNG seed. We generate an assignment list (CSV) with columns: `id`, `stratum_id`, `block_id`, `arm`. The CSV and the RNG seed are frozen and archived before outcome data are analyzed.

Allocation Concealment and Integrity

Randomisation scripts are versioned; assignment is applied automatically at survey start using the pre-generated list. We log assignment time, stratum, and block to enable ex-post audits.

8. Estimands and Synthetic Integration (Model-Assisted Only)

Estimands, Estimators, Weights, Interpretation

Let $\mathcal{D}_H$ be human data and $\mathcal{D}_S$ validated synthetic data. Per-observation weight

$$w_i = \begin{cases} 1, & i \in \mathcal{D}_H, \\ \omega, & i \in \mathcal{D}_S, \end{cases} \qquad \omega \in \{0, 0.2, 0.35, 0.5, 0.7\}.$$

Confirmatory estimand ($\omega = 0$). Human-population ATE on Y_1 via post-stratified difference-in-means:

$$\hat{\tau}(0) = \sum_s \pi_s \Big(\bar{Y}_{1s}^{(T=1)} - \bar{Y}_{1s}^{(T=0)} \Big),$$

with HC2 robust SEs clustered by stratum.
Augmented estimators ($\omega > 0$).

$$\hat{\tau}(\omega) = \sum_s \pi_s \left(\bar{Y}_{1s}^{(T=1,w)} - \bar{Y}_{1s}^{(T=0,w)} \right),$$

interpreted as a mixture-population estimand governed by ω. We report both $\omega = 0$ and $\omega > 0$ with clear labeling.

Human Pilot ("Small Pilot")

Size/selection. $N_{\text{pilot}} = 120$, balanced by primary strata using quotas.
Uses. (a) derive empirical bands for means/variances by stratum×arm; (b) estimate ATE *direction* with wide CI; (c) tune temperature grid and rejection bandwidths.
Bias control. Pilot sets tolerances only; it does not encode magnitude priors. The main $N = 300$ sample is independent; pilot respondents are not reused.

Choice of ω (Pre-Specified)

Split the main human sample into train/validation (60/40), stratified. On train: fit CAP and run validations. On validation: compute RMSE between augmented and human-only ATEs across bootstrap splits; select $\omega \in \{0.2, 0.35, 0.5, 0.7\}$ minimizing RMSE *subject to* equivalence (± 0.15 SD). Tie-break to smaller ω.

9. Analysis

Primary and Secondary Models

Primary confirmatory: $\omega = 0$ post-stratified difference-in-means for Y_1; HC2 SEs clustered by stratum.
Secondary: Y_2 (logit with stratum FE; AME reported); Y_3 (OLS; ordinal logit as robustness).
HTE: interactions for ideology (3), education (3), gender (2) with Holm–Bonferroni adjustment; exploratory HTE for age/income/region with BH-FDR.

Equivalence to Human-Only (TOST)

We conduct TOST comparing $\hat{\tau}(\omega)$ and $\hat{\tau}(0)$ with margin ± 0.15 SD (SD computed from human-only pooled variance aligned to post-stratification). Augmented estimates are precision-acceptable only if equivalence holds and sensitivity checks pass.

Missing Data, Non-Compliance, Attrition

Missing: MICE (20 imputations) with outcomes, treatment, and strata; Rubin's rules; imputation diagnostics reported.
Non-compliance: ITT primary; CACE via 2SLS using assignment as instrument when exclusion restriction is plausible (logic documented).
Attrition: stabilized IPW from completion propensity within strata; weights truncated at 1st/99th percentiles as robustness; report IPW and complete-case.

10. Implementation

Data Management and Reproducibility

We freeze stimuli, prompts, seeds, and CAP configs before main analysis; archive commit hashes. Synthetic-generation logs: model, provider, version/date, context window, temperature, seeds, generation time, and validation pass/fail.

Code Organization and Automation

A single YAML config controls ω and model selection. Scripts/notebooks produce preregistered tables/figures end-to-end. Repository structure (illustrative):

```
/data_raw   (pilot.csv, main_human.csv, syn_model*.csv)
/data_clean
/src        (stats helpers: std_diff, var_ratio, TOST, HC2, IPW, MICE)
/notebooks  (00_data_checks, 10_validation, 20_confirmatory, 30_augmented)
/outputs    (tables, figs, validation_results.json)
/pap        (this TeX and compiled PDF)
```

Public Release and Ethics

Release de-identified data, code, and a Quarto/R/Python notebook; suppress/merge small cells ($n < 10$). IRB approval prior to fielding; consent discloses AI-assisted analysis. Synthetic data are never presented as human without explicit labeling.

Timeline and Milestones

Pilot (N=120) $\rightarrow$ derive bands and tune CAP $\rightarrow$ lock prompts/seeds;
Validation checks per model with pass/fail logs;
Main RCT (N=300) with stratified randomization;

Analysis (confirmatory $\omega = 0$ and augmented variants) and sensitivity suite;
Public release (registry ID, data/code, protocol logs).

11. Decision Rules (Pre-Registered)

1) If any model fails any validation check (Pattern Replication, Effect-Direction Coherence, or External Benchmarking), exclude that model's synthetic data from the main augmented analysis.
2) If fewer than two models pass all validation checks, label augmented estimates as *sensitivity only*; if no model passes, report *human-only*.
3) Choose ω by validation RMSE subject to equivalence (± 0.15 SD) relative to $\hat{\tau}(0)$; tie-break to smaller ω.
4) Sparse-cell rule (pre-specified): within a stratum family, first merge by nearest ideology bin; if still sparse, merge by education; document all merges.
5) Multiple testing: primary HTE families use Holm–Bonferroni (FWER control); exploratory families report BH-FDR.
6) Any deviation from this PAP will be versioned with a timestamped change log and justified in the final write-up.

References

A. Notation and Identification (Non-Estimand-Changing)

Let $Y_i(t)$ be potential outcomes for unit i under treatment $t \in \{0, 1\}$. The ATE is $\mathbb{E}[Y(1) - Y(0)]$. Under stratified randomization, $\{Y(1), Y(0)\} \perp\!\!\!\perp T \mid S$ with known $\mathbb{P}(T = 1 \mid S)$. Post-stratified difference-in-means is unbiased for the human ATE. Synthetic observations, when included, are explicitly down-weighted and admitted only after validation.

B. Power Calculation Sketch

Under a balanced two-arm stratified design, the detectable standardized effect is approximately

$$\delta^\star \approx z_{1-\alpha/2}\sqrt{\frac{2(1-\rho)}{N}} + z_{1-\beta}\sqrt{\frac{2(1-\rho)}{N}},$$

where ρ is the fraction of variance explained by strata. With $N = 300$ and modest ρ, $\delta^\star \in [0.22, 0.25]$ is feasible.

C. Analysis Outputs (Pre-Specified Tables/Figures)

- **T1:** Balance by arm within strata (human-only).
- **T2:** Primary effect on Y_1 ($\omega = 0$); augmented variants with TOST equivalence.
- **T3:** Secondary outcomes Y_2, Y_3 (robust CIs).
- **T4:** HTE in primary strata (FWER-controlled).
- **F1:** $\hat{\tau}(\omega)$ path for $\omega \in [0, 0.8]$.
- **F2:** Variance ratios and design effects by ω.
- **T5:** Multi-model sensitivity (per-model and pooled).

D. Stimuli and Prompts (to be archived)

We archive: (i) gain/loss stimuli text; (ii) persona prompt templates; (iii) task prompts; (iv) CAP configuration (temperature grid, rejection-sampling limit K); (v) seeds for generation and randomization; (vi) model names, versions, and dates.

E. Data Dictionary (Human and Synthetic)

We provide variable names, types (numeric, factor), coding (including factor levels for strata), missingness codes, and construction steps for composite scores (e.g., knowledge).